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Graded Assignment • 20 min

Due Sep 10, 11:59 PM IST

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

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1. (**Special note:** Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file here. If unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.)

   1 / 1 point

   Consider the following nonlinear program

   $$\begin{array}{ll} \min & (x_1-4)^2+(x_2-2)^2 \\ \text{s.t.} & 2x_1+x_2 \leq 6. \end{array}$$

   Let $f(x_1,x_2)=(x_1-4)^2+(x_2-2)^2$ be the objective function. What are the leading principal minors of the Hessian matrix of $f(x_1,x_2)$?

   - ○ 0 and 2.
   - ○ 0 and 4.
   - ○ 2 and 2.
   - ◉ 2 and 4.
   - ○ None of the above.

   Correct

2. Continue from the previous question. For the Lagrangian

   1 / 1 point

   $$\mathcal{L}(x_1,x_2|\lambda)=(x_1-4)^2+(x_2-2)^2+\lambda(6-2x_1-x_2),$$

   which of the following statements is correct?

   - ○ We should have $\lambda \geq 0$.
   - ◉ We should have $\lambda \leq 0$.
   - ○ We should have $\lambda > 0$.
   - ○ We should have $\lambda < 0$.
   - ○ None of the above.

   Correct

3. Continue from the previous question. According to the first-order condition of the Lagrangian, which of the following is a necessary condition for an optimal solution?

   1 / 1 point

   - ○ $x_1+2x_2=0$.
   - ◉ $x_1-2x_2=0$.
   - ○ $2x_1-x_2=0$.
   - ○ $2x_1+x_2=0$.
   - ○ None of the above.

   Correct

4. Continue from the previous question. Which of the following is a local optimal solution to the nonlinear program?

   1 / 1 point

   - ○ $(4,2)$.
   - ◉ $(\frac{12}{5},\frac{6}{5})$.
   - ○ $(3,0)$.
   - ○ $(0,6)$.
   - ○ None of the above.

   Correct

5. For the following statements, select all that are correct:

   1 / 1 point

   - [x] For a linear program, linear programming duality and Lagrange duality is equivalent (i.e., the dual programs obtained through the two ways are identical).

     Correct

   - [x] For an unconstrained nonlinear program, the KKT condition is equivalent to the first-order condition.

     Correct

   - [x] For an unconstrained nonlinear program, the KKT condition is necessary and sufficient.

     Correct

   - [ ] For a constrained nonlinear program, Lagrange duality provides a tight bound.
   - [ ] None of the above.